

Purine-induced alterations of dopamine metabolism in rat pheochromocytoma PC12 cells

D. A. Loeffler,^{1,2*} D. M. Camp,² P. L. Juneau,³ E. Harel⁴ and P. A. LeWitt^{2,5}

¹Department of Immunology and Microbiology, Wayne State University School of Medicine, Detroit, MI, USA; ²Department of Neurology, William Beaumont Hospital, Royal Oak, MI, USA; ³Biostatistics, Discovery, and Early Development Projects, Warner-Lambert Company, Ann Arbor, MI, USA; ⁴Clinical Neuroscience Program, Sinai Hospital, Detroit, MI, USA; and ⁵Departments of Neurology and Psychiatry, Wayne State University, Detroit, MI, USA

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ABSTRACT: Studies with cerebrospinal fluid from subjects with Parkinson's disease suggest that purine abnormalities may be present in this disorder. The effects of purines on dopamine metabolism have not been characterized, though adenosine is known to inhibit dopaminergic neurotransmission. In this study, dopamine, its precursor 3,4-dihydroxyphenylalanine (DOPA), and its degradation products 3,4-dihydroxyphenylacetic acid (DOPAC), and homovanillic acid (HVA) were measured in rat pheochromocytoma PC12 cells following 24-h incubation with 5, 50, and 500 μ M adenosine, adenine, guanosine, guanine, hypoxanthine, xanthine, and uric acid. Incubation with adenosine increased DOPA, DOPAC, and HVA, while adenine treatment decreased DOPA. Guanosine (500 μ M) decreased DOPA, dopamine, and DOPAC, while lower concentrations increased DOPAC and HVA. Incubation with guanine decreased dopamine, and xanthine decreased dopamine and DOPAC. Hypoxanthine and uric acid exerted minimal effects. These results indicate that purines exert a variety of effects on dopamine metabolism. The influence of purine metabolism on the dopaminergic deficit in the Parkinsonian brain merits further investigation. © 2000 Elsevier Science Inc.

KEY WORDS: Dopamine, Parkinson's disease, PC12, Purines.

INTRODUCTION

Adenine and guanine nucleotides and nucleosides perform multiple roles in the brain, including energy metabolism [38], "second messenger" signaling [2], neuromodulation [42], regulation of membrane receptors to external stimuli [3], and neurotrophic effects [14,15]. Abnormal purine metabolism in the brain has been reported in a variety of neurological disorders, including cerebral ischemia [21,26], depression [28], Huntington's disease [4], and Alzheimer's disease [7,39]. Cerebrospinal fluid studies in our laboratory [22] and elsewhere [39] have suggested that abnormal central nervous system (CNS) purine metabolism may also be present in Parkinson's disease (PD). PD is characterized by marked depletion of striatal dopamine (DA) due to loss of nigrostriatal dopaminergic neurons. A compensatory increase in DA synthesis and metabolic deg-

radation ("DA turnover") occurs in the surviving dopaminergic neurons [19]. We recently reported that, in the rabbit striatum, increased DA turnover induced by administration of either levodopa or reserpine results in decreased guanosine and increased guanine concentrations [24]. Whether the elevated DA turnover in the PD striatum induces similar alterations in purine metabolism is unknown. Little is also known of the influence of purines on DA metabolism, though the inhibitory effects of adenosine on DA neurotransmission have been well described [10,16,20,25,27]. A1 and A2a receptors in the striatum interact in an antagonistic fashion with DA D1 and D2 receptors, respectively [8,9,23]; the A1 receptor is inhibitory, while the A2a receptor is stimulatory [35]. Interestingly, A2a receptors in the striatum are present primarily on intrinsic neurons, presumably post-synaptic to dopaminergic terminals [1,35], and may not be present on dopaminergic neurons [43]. The objective of the present study was to investigate the effects of various purines on DA metabolism in an *in vitro* model of dopaminergic neurons, the PC12 cell line. PC12 is a sympathetic-like, pre-neuronal cell line derived from a transplantable rat adrenal medullary pheochromocytoma [11], which exhibits typical neuronal morphology after treatment with neuronal growth factor (NGF) [12]. PC12 cells synthesize and secrete DA [33,34], and metabolize DA via monoamine oxidase [41].

MATERIALS AND METHODS

PC12 Cells

PC12 cells were obtained from Dr. Gregory Kapatos (Department of Psychiatry, Wayne State University, Detroit, MI). Cells were grown in RPMI medium 1640 with L-glutamine (Gibco BRL, Grand Island, NY, USA), supplemented with 10% horse serum (HS; Gibco), 5% fetal bovine serum (FBS; Irvine Scientific, Santa Ana, CA, USA), 100 units/ml sodium penicillin G, and 100 μ g/ml streptomycin sulfate (Penicillin-Streptomycin Solution; Irvine Scientific) (hereafter, "growth medium"). HS and FBS were heat-inactivated (56°C, 20 min) prior to use.

* Address for correspondence: David A. Loeffler, D.V.M., Ph.D., Research Institute Department of Neurology, William Beaumont Hospital, 3601 W. Thirteen Mile Rd., Royal Oak, MI 48073, USA. Fax: 248-551-5069.

Preparation of Purine Solutions

Stock solutions (1 mM) of adenosine, adenine, guanosine, guanine, hypoxanthine, xanthine, and uric acid (all from Sigma Chemical Co., St. Louis, MO, USA) were prepared. All but xanthine and guanine were dissolved directly in growth medium. Xanthine was first prepared as a 50 mM solution in 100 mM NaOH, then further diluted 1:50 in growth medium. Guanine was prepared as a 30-mM solution in 200 mM NaOH, then diluted 1:15 in growth medium to a concentration of 2 mM; 1 N HCl was added to adjust the solution to pH 7.8, the pH of the original culture medium. Additional growth medium was then added to achieve a final concentration of 1 mM. Appropriate vehicle solutions were prepared as above, but without purines. Purine stock solutions were sterile filtered (Millex-FG 0.2 μ m filter unit; Millipore Corp., Bedford, MA, USA), then further diluted in growth medium to obtain 5 μ M, 50 μ M, and 500 μ M concentrations.

Incubation of PC12 Cells with Purines

Log-phase PC12 cells were suspended in growth medium to 10^5 live cells/ml. One ml of cells was dispensed in each well of 24-well, flat bottom tissue culture plates (Becton Dickinson and Co., Lincoln Park, NJ, USA). After incubation for 48 h, medium was removed from each well, then 1 ml of each purine or vehicle solution (after brief warming at 37°C) was added to individual wells. Plates contained six wells each of cells incubated with 5 μ M, 50 μ M, and 500 μ M purine solutions, as well as six wells with the appropriate vehicle control. Incubation was continued for 24 h, then cells were harvested as described below.

Cell Harvesting

Cells were removed from each well by scraping. The contents of each well (cells plus medium) were collected, then placed in 1.5-ml tubes on ice. The tubes were centrifuged ($250 \times g$, 10 min, 4°C), then supernatant was removed and 500 μ l of cold 0.05 N perchloric acid (HClO₄) with 100 μ M sodium metabisulfite and 100 μ M ethylenediaminetetraacetic acid (EDTA) (all from Sigma) were added to each pellet. Cells were resuspended by vortexing, held briefly on ice, then sonicated on ice (five times at maximum, i.e., 70% power output, for 5 s each) with a Microson XL-2005 ultrasonic cell disruptor (Medsonic, Inc., Farmingdale, NY, USA). Sonicates were centrifuged ($12,000 \times g$, 10 min, 4°C), then filtered and stored at -70°C.

Measurement of DA and Related Compounds

DA, its precursor 3,4-dihydroxyphenylalanine (DOPA), and its metabolites 3,4-dihydroxyphenylacetic acid (DOPAC) and homovanillic acid (HVA) were assayed by high performance liquid chromatography with electrochemical detection, using procedures adapted from Robinson and Camp [37]. Separation of these compounds was achieved on a Microsorb C-18 column (5 μ m particles, 250×4.6 mm; Rainin Instruments, Woburn, MA, USA) maintained at a constant temperature of 29°C. Samples (20 μ l) were injected automatically onto the column using a refrigerated autosampler (ESA Model 540). Detection was accomplished using an ESA Coulochem II detector, an ESA Model 5021 conditioning cell (set at 10 mV), and an ESA Model 5011 dual analytical cell (with electrode 1 set at 450 mV and electrode 2 set at -400 mV). Under these conditions, DOPA, DOPAC, and HVA were quantified at electrode 1 and DA was quantified at electrode 2. The mobile phase consisted of 60 mM sodium phosphate, 30 mM citrate, 25 μ M EDTA, 10 mg/liter sodium dodecyl sulfate, 10% acetonitrile, and 5% methanol, and was adjusted to a final pH of 3.4. The concentration of each compound was determined by

TABLE 1
CONCENTRATIONS OF DOPAMINE (DA) AND RELATED
METABOLITES IN VEHICLE-TREATED PC12 CELLS

Experiment	DA	DOPA	DOPAC	HVA
1	27.09 $\pm$ 0.95	2.12 $\pm$ 0.13	0.94 $\pm$ 0.06	1.30 $\pm$ 0.11
2	79.44 $\pm$ 3.87	4.07 $\pm$ 0.19	2.62 $\pm$ 0.11	1.52 $\pm$ 0.08
3	85.06 $\pm$ 2.05	Not measured	2.89 $\pm$ 0.11	3.43 $\pm$ 0.17
4	31.27 $\pm$ 0.89	1.97 $\pm$ 0.08	0.70 $\pm$ 0.02	1.20 $\pm$ 0.06

Data are ng of each metabolite per well (equivalent to ng/ml) expressed as mean $\pm$ SEM. DOPA, 3,4-dihydroxyphenylalanine; DOPAC, 3,4-dihydroxyphenylacetic acid; HVA, homovanillic acid.

comparing peak heights with those of standards run with each experiment.

Statistics

This experiment was performed on four separate occasions, though some purine metabolites were omitted in some experiments. For each purine, pooled data from two to four experiments were analyzed. Concentrations of DOPA, DA, DOPAC, and HVA in purine-treated PC12 cells were expressed as percentages of their mean concentrations in cells incubated with the appropriate vehicle solutions. Differences in the concentrations of each DA-related compound in PC12 cells following incubation with vehicle solution, 5 μ M, 50 μ M, and 500 μ M of each purine were compared by Dunnett's T3 or C procedures [18]. Dunnett's T3 procedure was used when the total sample size for a comparison was less than 50; Dunnett's C procedure was employed for sample sizes of 50 or more. Both procedures maintain adequate statistical power for comparison of means and adjust for multiplicity of statistical testing, even in the presence of unequal group variances. All statistical comparisons were conducted at the 0.05 level.

RESULTS

Mean values for DOPA, DA, DOPAC, and HVA concentrations in lysates from vehicle-treated (control) PC12 cells are shown in Table 1 for the four experiments from which data were pooled. Metabolite concentrations were two- to three-fold higher in experiments 2 and 3 than in experiments 1 and 4, with the exception of HVA in experiment 1. This interassay variation necessitated normalization of data from purine-treated cells, relative to mean concentrations of DA and related metabolites in vehicle-treated cells, in each experiment in order to permit analysis of pooled data.

Adenosine

DOPA concentrations increased nearly four-fold ($p < 0.05$) with 500 μ M adenosine, and approximately 60% with lower adenosine concentrations ($p < 0.05$ for 50 μ M adenosine), compared with cells incubated with vehicle alone. DA concentrations were unchanged. DOPAC and HVA tended to increase in adenosine-treated cells (both $p < 0.05$ vs. vehicle-treated cells for 50 μ M and 500 μ M adenosine) (Fig. 1).

Adenine

DOPA was decreased following treatment with all adenine concentrations, while DA and DOPAC were unchanged. HVA was

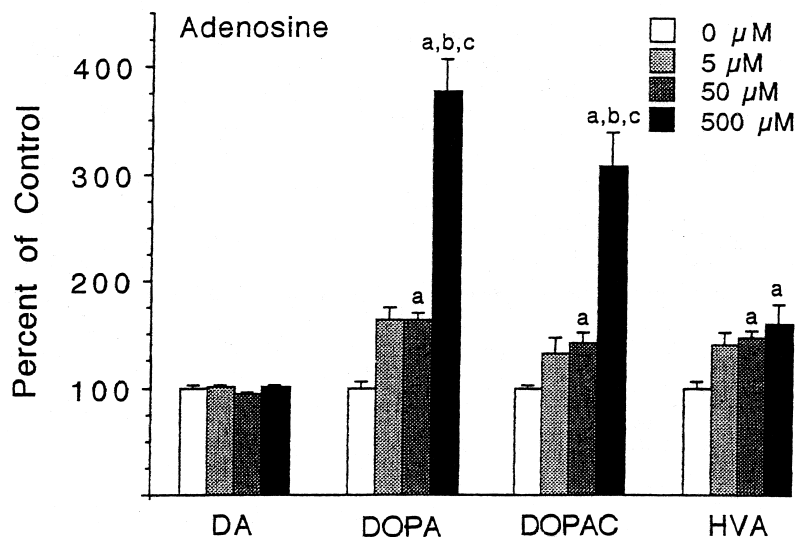


FIG. 1. Influence of adenosine on dopamine (DA) and related metabolites in PC12 cells. ^a $p < 0.05$ vs. 0 μ M; ^b $p < 0.05$ vs. 5 μ M; ^c $p < 0.05$ vs. 50 μ M. Abbreviations: DOPA, 3,4-dihydroxyphenylalanine; DOPAC, 3,4-dihydroxyphenylacetic acid; HVA, homovanillic acid.

decreased with 500 μ M adenine, but unchanged with lower adenine concentrations (Fig. 2).

Guanosine

DOPA and DA were decreased with 500 μ M guanosine, but were unchanged with lower guanosine concentrations. DOPAC was also decreased with 500 μ M guanosine. Conversely, DOPAC and HVA were increased in the presence of 5 μ M and 50 μ M guanosine (Fig. 3).

Guanine

DA was decreased with 50 and 500 μ M guanine, and DOPA was decreased with 500 μ M guanine. DOPAC and HVA were unchanged (Fig. 4).

Xanthine

DA and DOPAC were decreased following incubation with 500 μ M xanthine, while DOPA and HVA were unchanged (Fig. 5).

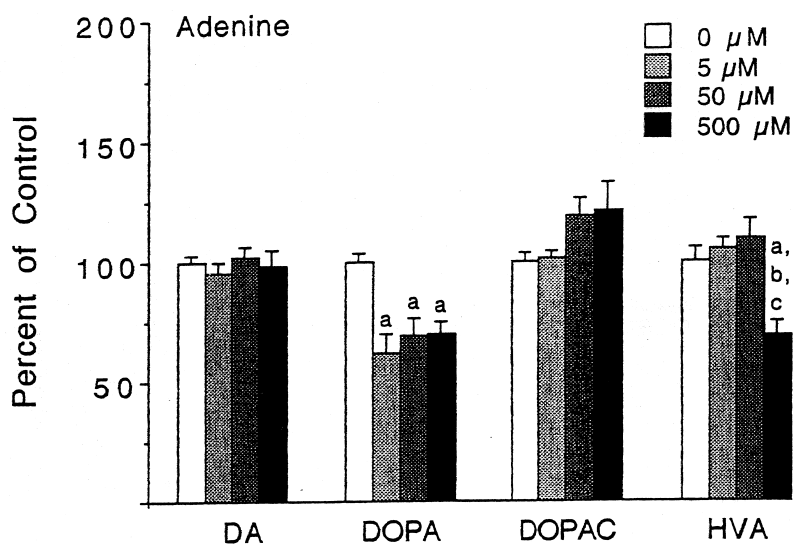


FIG. 2. Influence of adenine on dopamine (DA) and related metabolites in PC12 cells. ^a $p < 0.05$ vs. 0 μ M; ^b $p < 0.05$ vs. 5 μ M; ^c $p < 0.05$ vs. 50 μ M. Abbreviations: DOPA, 3,4-dihydroxyphenylalanine; DOPAC, 3,4-dihydroxyphenylacetic acid; HVA, homovanillic acid.

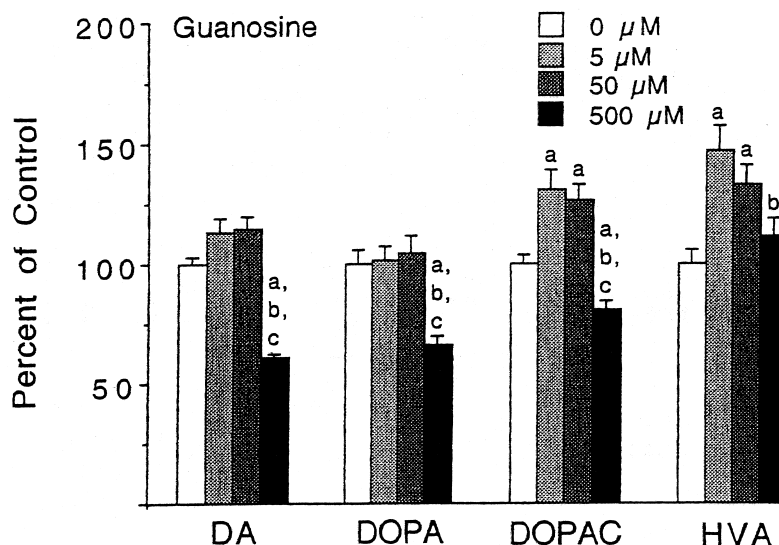


FIG. 3. Influence of guanosine on dopamine (DA) and related metabolites in PC12 cells. ^a $p < 0.05$ vs. 0 μ M; ^b $p < 0.05$ vs. 5 μ M; ^c $p < 0.05$ vs. 50 μ M. Abbreviations: DOPA, 3,4-dihydroxyphenylalanine; DOPAC, 3,4-dihydroxyphenylacetic acid; HVA, homovanillic acid.

Hypoxanthine and Uric Acid

No significant changes were detected for DOPA, DA, DOPAC, or HVA (data not shown).

DISCUSSION

The present study investigated alterations in intracellular concentrations of DA and related metabolites in PC12 cells following treatment with various purines. Treatment with adenosine pro-

duced an increase in DOPA, DOPAC, and HVA concentrations, while DA was unchanged. An increase in the DA degradation products DOPAC and HVA, without a concomitant decrease in DA concentration, suggests that release of DA from PC12 cells into the extracellular environment may have been impaired. Adenosine has been reported to inhibit release of DA *in vivo* [27], in striatal slices [16,20], and in striatal synaptosomes [25], but the mechanism by which adenosine might inhibit DA release in PC12 cells, as opposed to neurons, is unclear. Inhibition of DA release

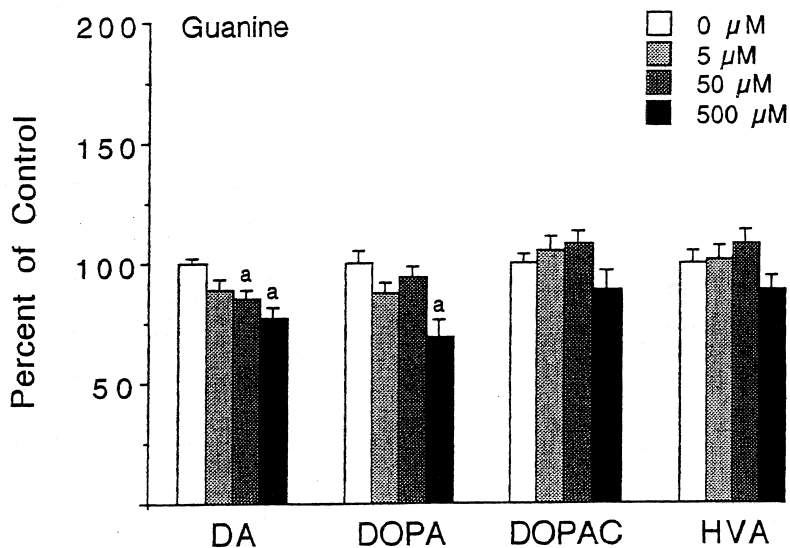


FIG. 4. Influence of guanine on dopamine (DA) and related metabolites in PC12 cells. ^a $p < 0.05$ vs. 0 μ M; ^b $p < 0.05$ vs. 5 μ M; ^c $p < 0.05$ vs. 50 μ M. Abbreviations: DOPA, 3,4-dihydroxyphenylalanine; DOPAC, 3,4-dihydroxyphenylacetic acid; HVA, homovanillic acid.

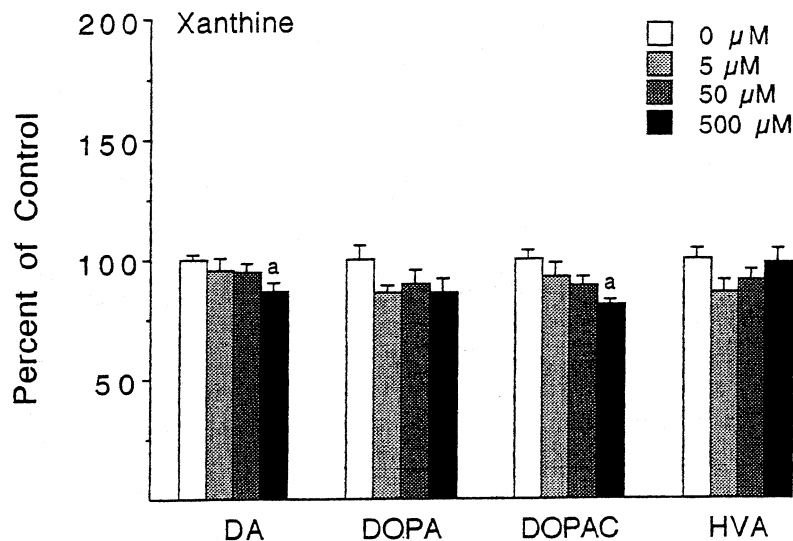


FIG. 5. Influence of xanthine on dopamine (DA) and related metabolites in PC12 cells. ^a $p < 0.05$ vs. 0 μ M; ^b $p < 0.05$ vs. 5 μ M; ^c $p < 0.05$ vs. 50 μ M. Abbreviations: DOPA, 3,4-dihydroxyphenylalanine; DOPAC, 3,4-dihydroxyphenylacetic acid; HVA, homovanillic acid.

has been reported following activation of adenosine A1 receptors [16,25,31], but PC12 cells are thought to lack these receptors [32]. In contrast, A2a receptors have been demonstrated on PC12 cells [17]. A2 receptor activation increases tyrosine hydroxylase activity [6,30], which converts tyrosine to DOPA; therefore, the increase in DOPA following adenosine treatment may have been due to activation of A2 (A2a) receptors. However, A2a receptors on PC12 cells rapidly desensitize following activation (i.e., in response to exposure to adenosine agonists [5]). Alternatively, because neurons possess bi-directional nucleoside transport systems [13], direct uptake of adenosine into PC12 cells may have occurred.

Guanosine has been shown to increase both *de novo* and NGF-stimulated neurite formation in PC12 cells [14,15]. The present study indicates that guanosine also influences PC12 DA metabolism. DOPA and DA concentrations decreased following incubation with 500 μ M guanosine. This concentration of guanosine also decreased DOPAC concentration, though lower guanosine concentrations resulted in an increase in both DOPAC and HVA. The significance of these results is unclear. Reduced DOPA and DA content suggests that guanosine treatment may have induced a decrease in DA synthesis, and reduced DOPAC suggests a concomitant decrease in DA degradation. The viability of PC12 cells, assessed by lactate dehydrogenase release, was similar after 24-h incubation in these purines and in vehicle solutions (data not shown), but the possibility cannot be ruled out that toxic effects of 500 μ M guanosine might account, in part, for the differences between 500 μ M and lower guanosine concentrations on PC12 DA metabolism. Like adenosine, guanosine levels markedly increase after focal CNS injury [40]; whether guanosine concentrations are also altered in PD and other progressive neurodegenerative disorders is unknown. The mechanism by which guanosine altered PC12 DA metabolism is unknown, because guanosine-specific receptors have not been described on PC12 cells (nor, for that matter, on neurons). Uptake of guanosine via the nucleoside transporter may have occurred. However, in an earlier study, the effects of guanosine on PC12 cell neurite outgrowth were not blocked by nucleoside transport inhibitors [14].

Alterations in PC12 DA metabolism were also detected following incubation with adenine, guanine, and xanthine. All concentrations of adenine decreased DOPA levels, and HVA was decreased by 500 μ M adenine. Incubation with all concentrations of guanine tended to decrease DOPA ($p < 0.05$ for 500 μ M guanine) and DA ($p < 0.05$ for 50 μ M and 500 μ M guanine), while xanthine treatment tended to decrease DA and DOPAC (both $p < 0.05$ for 500 μ M xanthine). These results suggest that these purines may also influence DA metabolism. Guanine can be converted to xanthine by guanine deaminase, but whether guanine's effects on DA metabolism were mediated in part through its conversion to xanthine is unknown.

In summary, purine nucleosides, purine bases, and related purine metabolites were found in this study to exert a variety of effects on DA metabolism in PC12 cells. Some purines induced consistent alterations in DA metabolism regardless of purine concentration, while the effects of other purines varied with concentration. Purine metabolism in the brain is complex, and purines may be readily interconverted by a variety of mechanisms, including intracellular metabolism and cellular ectonucleotidases. Elevated intracellular purine concentrations may also induce changes in other metabolic processes, which may in turn impact upon DA metabolism. Therefore the mechanisms by which the effects of the various purines on PC12 DA metabolism were mediated, and the significance of these *in vitro* results to the *in vivo* situation, remain to be determined. In view of the possibility that adenosine A2a receptor antagonists may be of value in treatment of PD [29,36], the influence of purine metabolism on the nigrostriatal DA deficit in this disorder merits further investigation.

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